

Linear Approximation and Differentials

Section 2.8

A. Differentials

$$\begin{array}{l} \text{Slope} = \frac{\Delta y}{\Delta x} \\ \text{Slope of the tangent line} = \frac{dy}{dx} \end{array} \quad \Rightarrow \quad \frac{\Delta y}{\Delta x} \approx \frac{dy}{dx}$$

$$\text{Also, } \frac{dy}{dx} = f'(x) \Rightarrow dy = [f'(x)] \cdot dx$$

Example: Find the differential of y given that:

1. (video) $y = 5x^3 \cdot \sin x$

2. (video) $f(x) = \sqrt{5x^2 + 2x - 12}$

3. $y = 5x^3$

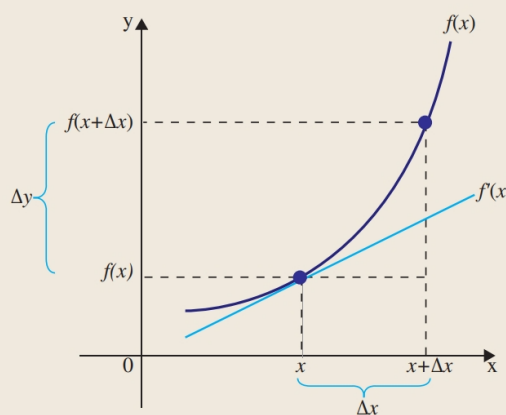
4. $y = \frac{3x^2 + 4x - 5}{5 \sin x + 2x}$

Note:

$$\Delta x \approx dx \text{ and } \Delta y \approx dy$$

$$dx \approx \Delta x = (x + \Delta x) - (x)$$

$$dy \approx \Delta y = f(x + \Delta x) - f(x)$$



5. **(video)** $A = \pi r^2$ (Area of a circle)

a. Use differentials to approximate the change in area when going from $r = 4$ ft to $r = 4.2$ ft. (Find dA)

b. **(video)** Find the actual change in area when going from $r = 4$ ft to $r = 4.2$ ft. (Find ΔA)

6. a. If $y = x^3$ then calculate Δy for $x = 2$ to $x = 2.01$

b. Calculate dy for $x = 2$ to $x = 2.01$

7. $V = \frac{4}{3}\pi r^3$ (volume of a sphere)

a. Use differentials to approximate the change in volume when going from $r = 3\text{ ft}$ to $r = 2.8\text{ ft}$

(Find dV : Start by finding $\frac{dV}{dr}$)

b. Find the actual change in volume when going from $r = 3\text{ ft}$ to $r = 2.8\text{ ft}$

(Find ΔV)

B. Linearization

Definition: The linearization of a function $f(x)$ at a fixed point a is given by the formula

$$L(x) = f(a) + f'(a)(x - a)$$

Slope of the tangent line =
 $m_T = f'(a)$

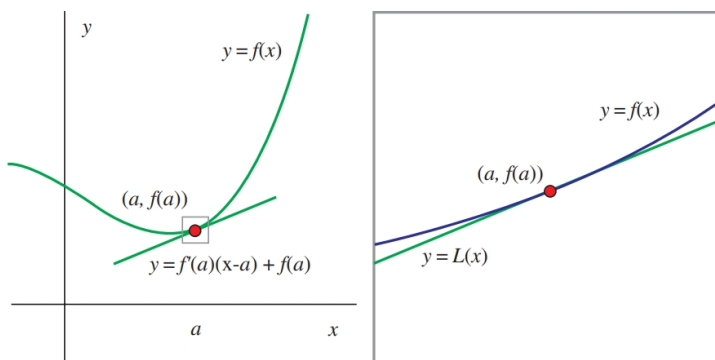
Point Slope:

$$(y - y_0) = m_T(x - x_0)$$

$$(y - f(a)) = f'(a)(x - a)$$

$$y = f(a) + f'(a)(x - a)$$

$$L(x) = f(a) + f'(a)(x - a)$$



Examples

8. (video) Find the linear approximation of $f(x) = \sqrt{x^2 + 3}$ at $a = 2$

9. (video) Use linearization techniques to approximate $\sqrt{25.1}$

10. Find the linear approximation of $f(x) = \cos(5x)$ at $a = \frac{\pi}{2}$

11. Use linearization techniques to approximate $\sqrt{16.1}$

12. Find the linear approximation of $f(x) = \sqrt{4-x}$ at $a = 0$ and use it to approximate $\sqrt{3.9}$ and $\sqrt{4.1}$.

13. Use a linear approximation to approximate 2.001^6 as follows:

The linearization $L(x)$ to $f(x) = x^6$ at $a = 2$ can be written in the form $L(x) = mx + b$. Using this, the approximation for 2.001^6 is

- 14.** The edge of a cube was found to be 60 cm with a possible error of 0.5 cm. Use differentials to estimate the following:
- a.** The maximum possible error in the volume of the cube
 - b.** The relative error in the volume of the cube
 - c.** The percentage error in the volume of the cube